Supplementary Tables

Title: Bilingual Retrieval-Augmented Access to ANVISA Regulatory Documents

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Supplementary Table S1. Expanded development decision table

Question category or system issue	What the system did well	What went wrong	What was changed	What this means
Source cleaning and document identity	The project identified the intended regulatory corpus.	Local source files included formatting artifacts and one misleading local file name.	Source text was cleaned and document identity was checked against official regulation names.	Regulatory RAG needs a stable and auditable corpus before retrieval metrics are interpretable.
Direct article questions	Retrieved and cited the correct article for compact questions.	Some answers still omitted conditions.	Article-level indexing and completeness prompting were added.	Useful for navigation, but citations must be checked.
English questions over Portuguese sources	Improved after Portuguese retrieval anchors.	English completeness was initially weak.	Conservative bilingual term anchors were added.	Bilingual search is feasible but needs controlled terminology.
Rule-plus-exception questions	Often found the right document.	Sometimes gave the exception without the general rule.	Rule-plus-exception failures were tracked separately.	Regulatory answers must preserve legal structure.
Long-list questions	Often found the right article.	Omitted one element, condition, or deadline.	Answer-completeness checks were added.	Average metrics can hide operationally important omissions.
False-premise questions	Could correct the premise and cite evidence.	Automated labels could confuse false premise with refusal.	False-premise questions were kept as a separate category.	These questions should not be merged with ordinary unanswerable questions.
Current or operational facts outside corpus	Refused after calibration.	Earlier system answered from nearby general text.	A stricter refusal rule and separate refusal test were added.	Refusal calibration is central for regulatory safety.
AI-authored prospective-test questions	Confirmed strong answerable-question behavior.	Refusal correctness dropped to 0.700.	Refusal was repaired using separate development data.	AI-authored prospective testing changed the project story.

Supplementary Table S2. Expanded representative answer outcomes

Question type	Example question	System behavior	Interpretation
Correct answer, pharmacovigilance	Which minimum information makes a notification valid and reportable?	Retrieved RDC 406/2020, Art. 32 and listed the minimum elements.	Correct source and correct main answer.
Correct false-premise handling	Is a notification valid without patient identification if the suspected medicine is named?	Answered no and cited RDC 406/2020, Art. 32.	Useful because it corrected the premise.
Correct clinical-trial safety answer	Does possible SUSAR reporting allow unblinding all participants?	Answered no; unblinding is limited to the affected participant and cited RDC 945/2024, Art. 67.	Correct and relevant to trial conduct.
Correct refusal	What is the current MedDRA version required today?	Refused because the indexed documents mention MedDRA but not the current required version.	Appropriate refusal for current operational information.
Correct refusal	What is the current phone number for Anvisa support for the adverse-event notification system?	Refused because the corpus does not contain current support contact information.	Appropriate evidence boundary.
Incomplete answer	What information must appear for non-printed instructions for use?	Retrieved RDC 751/2022, Art. 51 but omitted that instructions must remain available while the product is on the market.	Correct article, incomplete answer.
Incomplete answer	Can a foreign-manufactured biological product be registered without registration in the manufacturing country?	Gave the exception but did not clearly state the general rule.	Risk of treating an exception as the whole rule.
False refusal	Under RDC 945/2024, until when must DSURs be sent to Anvisa and what is the annual reference date?	Refused in one English prospective-test row although the corpus contained the answer.	Retrieval or answer threshold failure.
Refusal failure before calibration	What is the current electronic form model for notifying SUSARs to Anvisa?	Earlier AI-authored prospective testing sometimes answered with the general electronic-notification requirement.	Related context was mistaken for sufficient evidence.

Supplementary Table S3. Evaluation-stage summary

Stage	Main finding	Use in interpretation

Initial held-out evaluation	English completeness was the weak point; some answers omitted list items, conditions, or exceptions.	Development diagnosis.
Post-repair internal evaluation	High internal scores showed that the repaired system passed the curated benchmark.	Internal technical check, not independent validation.
AI-authored prospective stress test	Answerable questions were mostly supported, but refusal behavior was weak.	Diagnostic prospective evidence that changed the project focus.
Refusal-focused AI-authored test	Refusal improved after separate calibration, with no over-refusal on answerable controls.	Focused AI-authored check of refusal calibration.

Supplementary Table S4. Main-table disposition after revision

Original table	**Revision decision**	**Reason**
Table 1. Regulatory domains represented in the corpus	Replaced with a shorter corpus-and-use-case table.	Reduces page splitting and overlaps less with Figure 2.
Table 2. Question categories and corrective actions	Condensed in the main text and expanded as Supplementary Table S1.	Full version was too dense for the main manuscript.
Table 3. Representative examples	Condensed in the main text and expanded as Supplementary Table S2.	Keeps the story visible while moving audit-level detail to the supplement.
Table 4. Summary of evaluation stages	Removed from the main text and moved to Supplementary Table S3.	Redundant with Figure 3A and Figure 3B.

Supplementary Table S5. Decision trace from course notebooks to final ANVISA RAG

Stage	**Local evidence**	**Main decision**	**Effect on final system**
Course notebook 1	Downloads/1_LLM.ipynb; notebooks/1_LLM.ipynb	Use Python project scaffold with environment, config, model loader, engine, and main loop.	Final repository kept separated configuration, model loading, engine, and app modules.
Course notebook 2	Downloads/2_chatbot.ipynb; notebooks/2_chatbot.ipynb	Add chatbot behavior and system prompt.	Final app kept an interactive question-answer interface.
Course notebook 3	Downloads/3_rag.ipynb; notebooks/3_rag.ipynb	Adapt simple RAG from generic text to selected ANVISA documents.	The corpus became closed and regulatory rather than generic.

Borrowed Shenia source	src_RAG_Shenia/engine.py	Evaluate semantic splitting, HyDE, reranking, and chat memory ideas.	Reranking was retained conceptually; the final system used article-aware regulatory indexing instead.
First ANVISA implementation	/Users/cmontefusco/ Coding_projects/Rag_project	Build ANVISA app and index ANVISA text files.	This recovered folder is the best local ancestor of the publication repository.
Publication pipeline	src/article_chunking.py; src/retrieval.py; src/prompts.py; evaluation/publication_eval.py	Add cleaning, article chunks, hybrid retrieval, reranking, bilingual anchors, citation/refusal rules, and AI judging.	The final claim became closed-corpus traceable information access, not regulatory advice.

Supplementary Table S6. Investigation of saturated internal metrics

Observation	**Why it was suspicious**	**Investigation**	**Reliability interpretation**
Internal refusal correctness was 1.000.	The benchmark and repairs were produced during development.	Ran an AI-authored prospective stress test once after freezing.	The internal value was not used as proof of general refusal reliability.
Post-repair values approached 1.000.	Repair sets were not paired repeats and followed error review.	Reported Figure 3A as development history.	These values show controlled repair of known failure modes.
Prospective faithfulness was also 1.000.	Faithfulness can saturate for short cited answers.	Reported completeness, citation correctness, retrieval, and examples beside faithfulness.	Faithfulness alone does not rule out incomplete answers.
Prospective refusal correctness was 0.700.	This contradicted internal refusal correctness of 1.000.	Reviewed failures involving related but insufficient context.	This was evidence that internal 1.000 metrics were too narrow.
Refusal-focused test reached 0.944.	A high post-calibration value can still be overread.	Reported answerable-control completeness of 0.908.	Refusal improved, but completeness trade-offs remain.